

Multiple Choice

1. **Answer:** D

Options: A) Power emitted is 1/16 times as high; peak emission wavelength is 1/2 as long.; B) Power emitted is 1/4 times as high; peak emission wavelength is 2 times longer.; C) Power emitted is 1/4 times as high; peak emission wavelength is 1/2 as long.; D) Power emitted is 1/16 times as high; peak emission wavelength is 2 times longer.

Note: Correct option: D - Power emitted is 1/16 times as high; peak emission wavelength is 2 times longer.

2. **Answer:** C

Options: A) Mass 5 kg weight 100 N; B) Mass 5 kg weight 50 N; C) Mass 5 kg weight 25 N; D) Mass 10 kg weight 100 N

Note: Correct option: C - Mass 5 kg weight 25 N

3. **Answer:** A

Options: A) one medium sized satellite and two small satellites.; B) no satellites.; C) one large satellite and three small satellites.; D) one large satellite.

Note: Correct option: A - one medium sized satellite and two small satellites.

4. **Answer:** D

Options: A) Mars does not have enough internal heat to drive the greenhouse effect; B) Mars is too far from the sun for the greenhouse effect to work; C) the greenhouse effect requires an ozone layer which Mars does not have; D) the atmosphere on Mars is too thin to trap a significant amount of heat

Note: Correct option: D - the atmosphere on Mars is too thin to trap a significant amount of heat

5. **Answer:** D

Options: A) Will the Sun become a black hole?; B) Is the universe infinitely large?; C) How old is the visible universe?; D) Are we alone in the universe?

Note: Correct option: D - Are we alone in the universe?

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'An idealized physical object that absorbs all electromagnetic radiation.'

7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'You can never prove your theory to be correct only "yet to be proven wrong".'
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The time it takes for the Moon to make one circuit of our sky is one solar day..
Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** In the inner part of the nebula only metals and rocks were able to condense because of the high temperatures whereas hydrogen compounds although more abundant were only able to condense in the cooler outer regions.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key